GTransCD: Graph Transformer-Guided Multitemporal Information United Framework for Hyperspectral Image Change Detection

Xiaoyang Zhao, Siyao Li, Tingting Geng, and Xianghai Wang

***Abstract*— Using multitemporal hyperspectral images (HSIs) to obtain fine-grained land cover change information is an essential task in remote sensing (RS) image processing. Convolutional neural networks (CNNs), which have strong feature extraction and nonlinear regression capabilities, have recently aided in the advancement of this subject. However, the performance of these supervised methods is usually limited by small receptive field and less labeled samples. To this end, how to break through the aforementioned bottleneck and build a more suitable change detection (CD) framework for HSI is a crucial and challenging issue. To this end, a graph transformer-guided multitemporal information united framework for HSI-CD (GTransCD) is proposed, which mainly consists of the following three components: 1) applying transformer to the graph structure, a salient relationship strengthening graph transformer (GTrans) module is created, making it possible for the network to capture distant change information, and on this basis, local- and global-range information are aggregated simultaneously; 2) a gated change information fusion (GCF) unit is designed to inject the GTrans-guided change features into the original bitemporal concatenated features to further enhance the representation of change information in the network; and 3) a general HSI-CD framework that can organically blend change features guided by GTrans module with original features is proposed, with the intention of reducing the reliance on training samples by utilizing the semi-supervised learning mode of graph neural networks. numerous experiments demonstrate the proposed GTransCD surpasses the state-of-the-art methods and has a high level even at low sampling rates. The source code of the proposed framework will be released at https://github.com/zxylnnu/GTransCD.**

***Index Terms*— Change detection, deep learning, graph transformer (GTrans), hyperspectral image (HSI), multitemporal analysis.**

I. Introduction

With the rapid development of spectral imaging technology, hyperspectral remote sensing (RS) not only obtains spatial information of ground objects but also captures continuous spectral information. The "integration of map and spectrum" nature of hyperspectral imagery provides a strong basis for distinguishing similar objects with different spectrums and different objects with similar spectrums. Hyperspectral image change detection (HSI-CD) can effectively monitor the dynamic changes and spatial differences of land surface by analyzing HSIs acquired at the same geographical location but at different times. It has become one of the research hotspots in the field of RS in recent years due to its application in many fields such as urban expansion, disaster monitoring, and map-making [1].

Early HSI-CD methods typically adhere to a unified detection framework, which primarily includes the subsequent three steps.

1) *Image Preprocessing:* Geometric registration, radiometric correction, and other operations are performed on multitemporal images to reduce pseudo-change information brought by differences in image representation.
2) *Similarity Measurement:* Use pixel-level algebra, feature-level transformation, or object-level classification method to generate the change map.
3) *Results Evaluation:* Compare the change map with the visually interpreted ground truth (GT) to assess the detection accuracy. Although the above-mentioned traditional methods are highly efficient, the detection results are often restricted by the threshold.

The rise of deep learning technology has brought new solutions to the intelligent processing of RS images, and CD methods have begun to explore the automation mode in recent years, as a result of the significant increase in computational power and an enormous amount of RS data driven by. One of the reasons why deep learning-based methods achieve high detection accuracy is because of this type of feature, which does not depend on the hand-crafting feature extraction experience, making the network automatically learns the essential features as the basis for detection. In particular, the unique local perception and parameter-sharing mechanism of convolutional neural network (CNN) can efficiently abstract concrete image data to high-level features step by step, demonstrating excellent detection performance and infinite potential. Luo et al. [2] propose a multiscale diff-changed feature fusion network (MSDFFN) to capture fine changes at each scale. Furthermore, a network structure known as LSTM is also frequently employed in HSI-CD to mine the temporal correlation between multitemporal images. According to Shi et al. [3], a multi-path convolutional LSTM neural

Manuscript received 14 October 2023; revised 4 November 2023; accepted 1 December 2023. Date of publication 4 December 2023; date of current version 13 December 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41971388 and in part by the Key Scientific Research Project of Liaoning Provincial Education Department under Grant JYTZD2023101. *(Corresponding author: Xianghai Wang.)*

Xiaoyang Zhao is with the School of Geographical Sciences, Liaoning Normal University, Dalian 116029, China.

Siyao Li and Tingting Geng are with the School of Computer and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China.

Xianghai Wang is with the School of Geographical Sciences, Liaoning Normal University, Dalian 116029, China, and also with the School of Computer and Artificial Intelligence, Liaoning Normal University, Dalian 116029, China (e-mail: xhwang@lnnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2023.3339247

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

network (MP-ConvLSTM) is proposed, which uses CNN and LSTM to extract multiscale spatiotemporal-spectral features of images in parallel; Qu et al. [4] explore the role of multilevel fusion strategy in the field of HSI-CD by proposing a multilevel encoder-decoder attention network (ML-EDAN) and use LSTM to analyze multilevel spatiotemporal-spectral association information. While the above CNN-based methods have achieved relatively satisfactory detection results, they are usually limited by the following factor: first, traditional CNN networks take hyperspectral patches as detection units, making the feature extraction comes within the fixed patch. It is difficult to improve the current region by learning about distant similar change information. To solve this problem, some researches take advantage of dilated and deformable convolutions to expand receptive information. Specifically, Song et al. [5] use dilated convolutions with different dilated ratios instead of conventional convolutions to expand the receptive field of the network; Yang et al. [6] use continuous dilated-deformable feature extraction blocks to acquire subpixel information flexibly beyond the sampling position. Although these methods attempt to increase the amount of information perceived by the network through different patterns, the source of the information is still from the current input patch. And the methods of large-size patch-cutting may introduce heterogeneous information.

In addition, the detection accuracy of these supervised methods also depends on a large number of training samples. However, due to the mixed pixel phenomena, interpreting HSI is more challenging. For this reason, to lessen the dependence of the network on training samples, Liang and Chen [7] suggest a hierarchical clustering network with the self-supervised learning strategy; inspired by intrinsic image decomposition, Sun et al. [8] design a two-stream coupled autoencoder using the unsupervised learning mode, which can not only alleviate the sample demand of the network but also achieve an advanced detection level by separating the effective spectral change components. It can be seen that, although the issue of inadequate samples has received extensive attention in this field, the study is not yet comprehensive.

In light of the aforementioned issues, some studies have recently applied graph theory knowledge to this field with the goal of using the semi-supervised learning mode to propagate the label to nodes with homogeneous attributes. Nevertheless, it is clear that graph-based methods suffer the following difficulties on the whole: on the one hand, there is a lack of fine HSI-CD framework under the condition of a few training samples; on the other hand, although the methods take the entire image as the input, both GCN and GAT can only aggregate neighbor node information into the current one. This local message-passing mode still cannot make use of distant information, without realizing the global interaction in a true sense. To address the aforementioned challenging problems, a graph transformer-guided multitemporal information united framework for HSI-CD (GTransCD) is proposed, which organically blends the change features guided by a salient relationship strengthening graph transformer (GTrans) module with the original bitemporal concatenated features, and is able to propagate labels to unlabeled samples in a semi-supervised learning mode by constructing the topological structured graph, reducing the reliance of the network on a large number of training samples. The main contributions are as follows.

1) Create a salient relationship strengthening GTrans module: by executing this module on the entire image via applying the transformer to the graph structure, it is possible for the network to capture distant change information, and on this basis, local- and global-range information are aggregated simultaneously to enhance the representation of current detection region.
2) Design the gated change information fusion (GCF) unit to inject the GTrans-guided change features into the original bitemporal concatenated features to further enhance the representation of change information in the network.
3) Propose a general HSI-CD framework that can organically blend change features guided by GTrans module with original features, aiming to take advantage of the semi-supervised learning mode of GNNs to fully utilize the hidden information contained in unlabeled samples and further reduce the reliance on a large number of training samples.

The remainder of this article is structured as follows: The application of GNN in HSI processing is introduced in Section II, along with transformer-based HSI-CD methods; the steps and detailed descriptions of the proposed GTransCD are provided in Section III; the evaluation indicators and visualization results of each experiment are presented in Section IV under the conditions of various sampling rates; finally, the conclusion is drawn in Section V.

II. Related Works

In this section, GNN is first briefly described and its application to HSI processing is presented; next, the use of transformer in the field of HSI-CD research is detailed.

A. GNN and Its Application to HSI Processing

Graph neural networks have drawn much interest for the exceptional capacity to model topologically structured data and have been widely employed in a variety of learning tasks, including recommendation systems, molecular property prediction, and so on. Recurrent graph neural networks (RecGNN), the first application of neural networks to directed acyclic graphs by Sperduti et al. [9], started the early research on GNNs. It is implemented by assuming that the target node in the graph can exchange information iteratively with its neighbor nodes under this framework until a stable equilibrium state is reached. However, the high-computational complexity of the learning process severely constrains the generalization of the framework to multiple applications.

Influenced by the great success of CNNs in RS image processing, a lot of research works have been devoted to redefining the convolutional operation acting on graph data, and are referred to as convolutional graph neural networks. They can be divided into two streams, the frequency domain-based methods and the spatial domain-based methods. Among them, frequency domain-based methods define graph

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

convolution by introducing filters from the perspective of graph signal processing. Classic frequency domain methods include Spectral CNN, ChebNet, GCN, and so on; the spatial domain-based methods inherit the idea of RecGNN and define the graph convolution through message propagation. More typical ones are gated graph neural networks, graph attention (GAT) networks, etc.

GCNs have recently been introduced to the HSI processing field and are demonstrating promising outcomes. To approximate convolution and fully utilize the spatial information of the current pixel, Qin et al. [10] use adjacent nodes in the graph to present spectral–spatial graph convolutional networks (S2GCNs); Mou et al. [11] create non-local graphs to enhance the embedding representation of spatial contextual information in GCNs; Wan et al. [12] mine multiscale spectral–spatial diversity correlations by establishing multiple input maps of various neighborhood scales to comprehensively take multiscale information contained in HSIs into account. To break the limitation of spatial topology modeling at the same scale, Liu et al. [13] propose a multilayer superpixel graph U-Net, which generates finer feature representations in a progressive fusion pattern from coarse to fine; to effectively exploit spatial–spectral joint information, Guo et al. [14] merge the multilayer graph residual module and spectral domain feature continuous learning module; to make up for the shortcoming that GCN cannot capture pixel-level spatial features, Liu et al. [15] organically combine CNN and GCN branches to enable feature learning in both small-scale regular regions and large-scale irregular regions, respectively, and generate complementary spectral and spatial features.

For HSI-CD, Wang et al. [16] introduce GAT into HSI-CD research field for the first time, and propose a dual-branch framework based on temporal-spatial joint GAT and complementary strategy (CSDBF). The framework uses the spatiotemporal joint GAT branch to enhance the propagation of change information, and at the same time, the parallel CNN blocks can correct the wrongly detected pixels to obtain greater detection performance; Qu et al. [17] introduce GCN into HSI-CD research field for the first time, and design a dual-branch difference amplification graph convolutional network (D^2AGCN), using GCN to efficiently propagate superpixel-level features and amplify multitemporal difference features.

As can be seen, the researches of GNN on HSI processing are still immature, and there has only been a certain amount of study accumulation in classification. But for CD, how to implement the mining and usage of multitemporal information in GNN-based methods, there is still much work to be developed, which is also one of the key and challenging points in this field.

B. Use of Transformer in HSI-CD

Transformer [18] is a network architecture that can stably capture context information in sequence data by stacking multiple self-attention modules and is widely used in RS image-processing fields [19].

In recent years, transformer-based HSI-CD methods have achieved advanced detection performance. A hybrid spatial–spectral convolutional vision transformer (SSViT) is designed by Shafique et al. [20], which enhances the network to distinguish real changed features by generating pseudo-samples. Wang et al. [21] proposed a joint spectral–spatial–temporal transformer (SST-Former) to improve the feature representation of the spectrum in the network, using a multi-head attention model to extract spectral, spatial, and temporal features sequentially. Ding et al. [22] introduce a pixel embedding module to encode spatial and temporal change information for a comprehensive analysis of change features. To reduce the redundancy of changed feature maps, Hu et al. [23] suggest an efficient multitemporal self-attention network (EMS-Net), which enhances the compactness of unchanged features from the perspective of loss function. Sun et al. [24] propose an adaptive spatial–spectral transformer CD network (STCD-Former), which uses a difference amplification module to strengthen the spectral difference features learned by the transformer. Huang et al. [25] present a parallel spectral-spatial attention (PS2A) network to optimize the fusion of multilevel features and change features to overcome the problem of insufficient use of hierarchical features.

Further, the studies that follow are dedicated to exploiting nonocal information via mining out-of-patch data. To address the problem of insufficient multitemporal feature utilization, Wang et al. [26] propose a tri-branch framework, where each branch uses a transformer as the backbone to extract local features from the current patch, multitemporal features at the same position from another temporal patch, and spatial features at different positions from the same temporal patch, respectively; to investigate the interactions between global features, Hu et al. [27] design a global multi-head interactive self-attention change detection network (GlobalMind), using the global axial segmentation strategy to deploy the transformer over the entire image.

It is clear that the acquisition of global change information can significantly improve the embedding representation of change features in the network, but generally speaking, the current work is still in its early stages, necessitating the further development of more efficient methods to capture global change information.

III. Methodology

In this section, we first introduce the overall structure of the proposed GTransCD, and then describe each component in detail, including superpixel-level graph construction, graph learning, GCF unit, and the loss function.

A. Overall Structure of GTransCD

Define bitemporal HSIs as $X_1 \in R^{H\times W\times B}$, $X_2 \in R^{H\times W\times B}$, and the corresponding GT map is $Y \in R^{H\times W}$, where B is the number of bands, H and W are the height and width of each band. AS the GTransCD shown in Fig. 1, the framework also receives the differential image X_3 as input. $X_3 = |X_2 - X_1|$, $X_3 \in R^{H\times W\times B}$ is used to make the network further analyze the difference features between bitemporal images. Unlike frameworks that take hyperspectral patches as input, GTransCD accepts the entire HSI as input.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

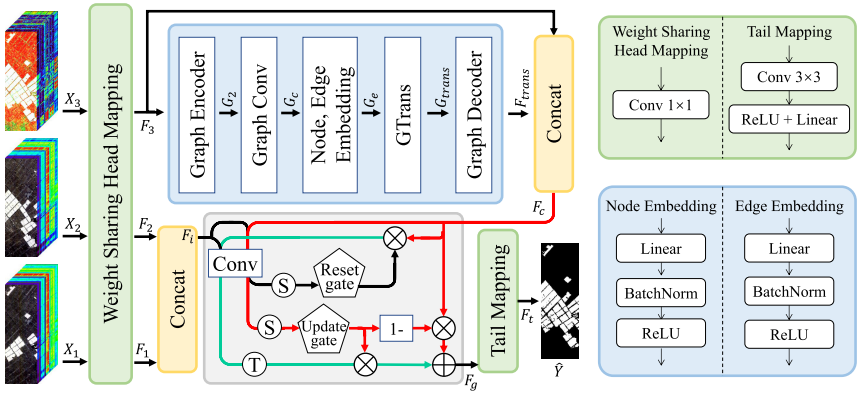


Fig. 1. Flowchart of the GTransCD. It mainly consists of four parts: weight sharing head mapping (colored green), graph learning (colored blue), GCF unit (colored gray), and tail mapping (colored green). The structure of some simple layers is shown on the right side.

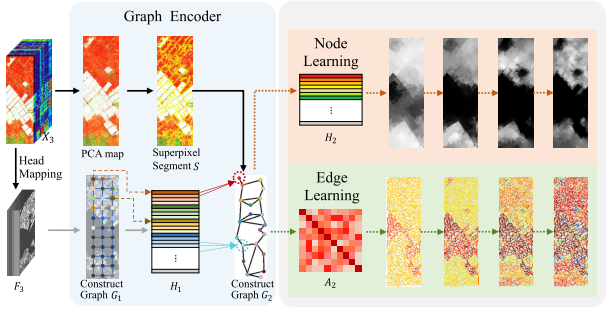


Fig. 2. Construction of superpixel-level graph.

First, GTransCD uses a weight sharing head mapping layer to perform simple dimensionality reduction and feature extraction on the three input HSIs. This layer only contains a vanilla 1×1 2-D convolution operation.

Second, before analyzing the change information contained in X_3, it is necessary to perform the preprocessing operation shown in Fig. 2 to construct graph, including PCA and superpixel segmentation. The specific modeling processing is described in Section III-B. The pixel-level graph of X_3 is first encoded into superpixel-level graph by graph encoder; and then, the graph features are updated by graph convolutional layer, node, and edge embedding layer; next, these features are input into the proposed GTrans module. It is able to fully capture the changed features in the X_3 to realize graph representation learning. This part is introduced in detail in Section III-C.

Finally, in the graph decoder layer, superpixel-level features are decoded into pixel-level features and they are then concatenated with X_3 features to obtain the whole change features. After that, these features are injected into the concatenated bitemporal features through the GCF unit to further enhance the representation of change information in the network, see Section III-D for details. The loss function is given at Section III-E.

The specific implementation steps of GTransCD are given in Algorithm 1.

Algorithm 1 Pipeline of GTransCD

Input: Hyperspectral images X_1, X_2, X_3; GT map Y; learning rate lr; training epochs E.
Output: Predict change map,$\hat{Y}$.

Employ PCA to X_3, noted as P.
Segment P into superpixels via Boruvka algorithm.
for $e = 1$ to E **do**
  Feature extraction and dimension reduction by weight sharing head mapping layer:
  $F_n = Conv_{1\times1}(X_n), n = 1, 2, 3$
  Construct pixel-level graph G_1 based on F_3.
  Construct superpixel-level graph G_2.
  Construct incidence matrix Q by Eq. 2.
  Encode G_1 to G_2 by Eq. 3∼4.
  Implement graph convolution to G_2 by Eq. 5∼6, noted as G_c.
  Implement node, edge embedding layer to G_c, noted as G_e.
  $G_e = Embed(G_c) = ReLU(BN(Linear(G_c)))$
  Implement GTrans module to G_e by Eq. 7∼8, noted as G_{trans}.
  Decode G_{trans} by Q^T, noted as F_{trans}.
  Concatenate F_{trans} and F_3, noted as F_c.
  Concatenate F_1 and F_2, noted as F_i.
  Input F_c and F_i into GCF unit by Eq. 9∼10, noted as F_g.
  Implement tail mapping layer to F_g, noted as F_t.
  $F_t = Linear(ReLU(Conv_{3\times3}(F_g)))$
  Predict the change map.
  $\hat{Y} = Softmax(F_t)$
  Calculate the loss L_{BCE} between $\hat{Y}$ and Y by Eq. 11.
  Back-propagate L_{BCE} and solve it with Adam optimizer.
end for

B. Construct Superpixel-Level Graph

To reduce the computational cost and obtain a structured graph that is more suitable for the boundary of ground objects, it is necessary to treat superpixels as analysis units. The specific process of constructing superpixel-level graph is shown in Fig. 2.

1) Superpixel Segmentation: First of all, to reduce the potential impact of noise and redundant information in X_3 on the subsequent segmentation, the top $p(p = 3)$ principal components in X_3 are extracted by PCA, denoted as P.

Then, assume that the acquired number of superpixels is $k(k \ll H \times W)$, the PCA map P can be divided into k superpixels, and each subregion is denoted as C_i $(i = 1, \ldots, k)$, so that, the superpixel segmentation $S = C_1 \cup C_2 \cup \cdots \cup C_k$.

Next, build a forest with $n(n = H \times W)$ trees, and calculate the edge weight between every two trees within the local four-neighborhood by l_1 distance.

Finally, use the Boruvka algorithm [28] to calculate the minimum spanning tree (MST) of the forest in bottom-up

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

manner. Find the neighbors connected by the lightest edge to the current tree. Following equation gives the expression of the shortest distance between two trees when C_2 is the nearest neighbor of C_1:

$$\mathcal{D}(C_1, C_2) = \min_{v_i \in C_1, v_j \in C_2, (v_i, v_j) \in \mathcal{E}} \left\| v_i - v_j \right\|_1 \tag{1}$$

where, v_i denotes the ith tree, $\mathcal{E}$ denotes the edges between each two nodes. Once $e_{i,j}$ is added to the MST, v_i and v_j will be connected and merged, and the number of trees in the forest will be reduced by 1, until there are k trees left.

2) Superpixel-Level Graph Construction: Before constructing the superpixel-level graph, define the feature map of X_3 after the weight sharing head mapping layer as $F_3 \in R^{H \times W \times B'}$, where B' represents the number of feature map channels. The undirected pixel-level local four-connected structured graph $G_1 = (V_1, E_1)$ shown in Fig. 2 is constructed based on F_3, which consists of n nodes $v_1 \in V_1$ and n^2 edges $e_1 \in E_1 \subseteq V_1 \times V_1$, where $n = \mid V_1 \mid = H \times W$, the corresponding node feature matrix $H_1 \in R^{n \times B'}$, and the ith row of H_1 represents the characteristics of node v_i.

Subsequently, define the undirected superpixel-level graph $G_2 = (V_2, E_2)$, which consists of k nodes $v_2 \in V_2$ and k^2 edges $e_2 \in E_2 \subseteq V_2 \times V_2$, and set the node feature matrix $H_2 \in R^{k \times d}$, adjacency matrix $A_2 \in R^{k \times k}$, d is the feature vector dimension of each node.

On this basis, to enable features propagate between pixel-level and superpixel-level, the incidence matrix $Q \in R^{n \times k}$ is defined, and the 2 gives the expression of Q. The physical meaning of this formula is: if node v_2^j contains node v_1^i, the value of matrix Q at position (i, j) is 1, and if node v_1^i does not belong to v_2^j, the value of $Q_{i,j}$ is 0

$$Q_{i,j} = \begin{cases} 1, & \text{if } v_1^i \in v_2^j \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Then, the node feature matrix H_2 of graph G_2 can be obtained according to the following equation:

$$\begin{cases} \hat{Q}_{i,j} = \dfrac{Q_{i,j}}{\sum_m Q_{m,j}} \\ H_2 = \hat{Q}_1^{\mathrm{TH}}. \end{cases} \tag{3}$$

Further, to enable the current node to perceive distant nodes and aggregate the global node features, the adjacency matrix A_2 is calculated as follows:

$$A_2^{i,j} = \exp\left(-\frac{\left\| H_2^i - H_2^j \right\|^2}{\sigma^2} \right) \tag{4}$$

where, the parameter σ is a constant and is set to 0.1, H_2^i, and H_2^j are the feature vectors of nodes v_2^i and v_2^j, respectively, and the higher the value of $A_2^{i,j}$, the node v_2^i is more similar with the node v_2^j.

C. Graph Learning

1) Graph Convolution: After obtaining the superpixel-level graph, use graph convolution first to update the node feature matrix H_2 and adjacency matrix A_2 to obtain graph G_c. Graph convolution can make the current node in the graph aggregate the information from other nodes, so that the spatial information and spectral information in the image can be spread. The update rule of adjacency matrix A_c of graph G_c is shown in the following equation:

$$A_c = S\left(\left(\tilde{H}_2 W_A \right) \left(\tilde{H}_2 W_A \right)^T \right) \odot A_2 + \lambda I \tag{5}$$

where, $S(\cdot)$ represents the Softmax normalization function, which is used to ensure the stability of feature propagation between related nodes and limit the feature propagation of unrelated nodes. $\tilde{H}_2 = \mathrm{BN}(H_2)$, $\mathrm{BN}(\cdot)$ is the batch normalization function, W_A is the parameter to be learned, which is acquired using the linear layer, $\odot$ represents the Hadamard product operation, λ is a learnable constant, and I is the identity matrix.

Furthermore, the node feature matrix H_c of graph G_c can be calculated according to the following equation:

$$H_c = \sigma\left(\hat{D}^{-\frac{1}{2}} A_c \hat{D}^{-\frac{1}{2}} \tilde{H}_2 W_H \right) \tag{6}$$

where, σ is a nonlinear activation function, the LeakyReLU$(\cdot)$ function is used here, $\hat{D}$ is the degree matrix of A_c, $\hat{D}^{i,i} = \sum_j A_c^{i,j}$, $\tilde{H}_2 = \mathrm{BN}(H_2)$, W_H is the parameter to be learned, obtained by a simple linear layer.

2) Salient Relationship Strengthening GTrans Module: Before using the GTrans module to obtain the affinity relationship between any two nodes, perform feature embedding on the graph G_c first (refer to Algorithm 1 for details), and obtain the graph $G_e = (V_e, E_e)$, which consists of $v_e \in V_e$ and $e_e \in E_e \subseteq V_e \times V_e$, with a node matrix $H_e \in R^{k \times \mathrm{ch}}$ and an adjacency matrix $A_e \in R^{k \times k}$, where k is the number of nodes, and ch represents the node-embedding feature dimension. The GTrans module structure is shown in Fig. 3. h_e^i is the embedding feature of the ith node in H_e, and $e_e^{i,j}$ is the intensity of the adjacency relationship between nodes v_e^i and v_e^j.

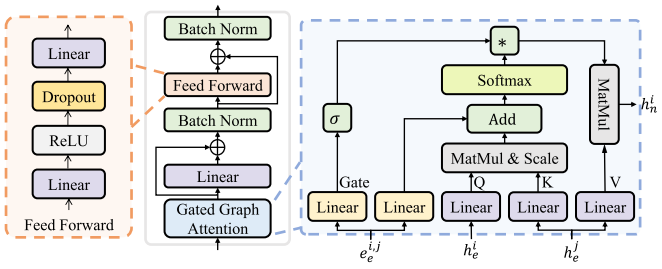


Fig. 3. Flowchart of the GTrans module. It mainly consists of five layers: gatedGAT (colored blue), linear, feed forward (colored orange), and two batch normalization layers.

The first layer in the GTrans module is the gated map attention layer, and the definition of this layer is shown in the following equations:

$$\mathrm{Attn}\left(h_e^i, h_e^j\right) = \mathrm{Softmax}\left(\frac{W_q h_e^i \left(W_k h_e^j \right)^T}{\sqrt{d_k}} + W_e e_e^{i,j} \right) \tag{7}$$

$$h_n^i = \mathrm{Attn}\left(h_e^i, h_e^j\right) * \sigma\left(W_g e_e^{i,j} \right) \times W_v h_e^j \tag{8}$$

where, "$*$" means element-wise multiplication, "$\times$" means matrix multiplication, W_q, W_k, W_v, W_e, W_g are all learnable parameters, which are learned by Linear$(\cdot)$ layers, $\sigma(\cdot)$ is a nonlinear activation function, Sigmoid$(\cdot)$ is used here to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

further realize whether the information is propagated and the magnitude of the propagation. It can be seen that the GTrans module considers the adjacency relationship when learning the relationship between nodes, and considers the characteristics and relationships of each node more comprehensively.

Then, to deepen the depth of the module and enhance the feature representation, one linear layer followed by a BN layer and a feedforward layer followed by a BN layer are appended. And the feedforward layer includes a linear layer, ReLU(·), and Dropout. And on this basis, the residual connections are added.

D. GCF Unit

To decrease the potential risk of misclassification caused by superpixel segmentation and further highlight the difference information in bitemporal HSI images, the GCF unit injects the change features guided by GTrans into the original bitemporal concatenated features through the designed "reset gate" and "update gate." The reset gate can filter out the information that is invalid for the network and discard it; the update gate determines which information can continue to propagate in the network. The definitions are shown as follows:

$$\mathrm{RG},\ \mathrm{UG} = S(\mathrm{Split}(\mathrm{Conv}_{1\times1}(\mathrm{Concat}(F_i,\ F_c)))) \tag{9}$$

where RG is the reset gate, UG is the update gate, and $S(\cdot)$ represents the Sigmoid activation function. Then, the overall steps of the GCF unit can be expressed as

$$\begin{cases} F_h = \mathrm{Tanh}(\mathrm{Conv}_{1\times1}(\mathrm{Concat}(\mathrm{RG}\times F_c,\ F_i))) \\ F_g = (1-\mathrm{UG})\times F_c + \mathrm{UG}\times F_h. \end{cases} \tag{10}$$

E. Loss Function

After performing the tail mapping layer with convolution, ReLU and linear operations on F_g, the change map $\hat{Y} \in R^{\mathrm{HW}\times2}$ can be predicted using Softmax. For GTransCD, naive binary cross-entropy is used as the loss function to calculate the difference between the predicted change map and the GT map

$$L_{\mathrm{BCE}} = -\sum_{i=1}^{N}\left(Y_i \log \hat{Y}_i + (1-Y_i)\log\left(1-\hat{Y}_i\right)\right) \tag{11}$$

where, N is the total number of samples, Y_i is the ith element of the GT map, and $\hat{Y}_i$ is the ith element of the predicted change map. And use the Adam optimizer to train the network parameters to continuously reduce L_{BCE} until the network training epochs reach the maximum.

IV. Experiments and Analysis

Two sets of experiments—one with low sampling rates and the other with the typical sampling rates used in most research—are designed to test the effectiveness and excellent detection performance of the framework.

TABLE I
Dataset Details

Dataset	River	Farmland	USA
Size	463×241	450×140	307×241
Band	198	155	154
Date T_1	May 3rd, 2013	May 3rd,2006	May 1st,2004
Date T_2	Dec 31st, 2013	Apr 23rd,2007	May 8th,2007
Changed samples	9698	44723	16676
Unchanged samples	101885	18277	57311

A. Datasets and Experimental Setup

In this article, three commonly used HSI-CD datasets are employed to evaluate the detection performance of GTransCD, namely River [29], Farmland [30], and USA [31]. For each dataset, a pair of bitemporal HSIs and a corresponding binary GT map are included. These three datasets are described in depth in Table I, and the pseudo-color visualizations and the GT maps are displayed in Fig. 4.

This experiment is implemented under the PyTorch framework with an NVIDIA RTX 3090 GPU for training. Adam is selected as the network optimizer, the learning rate is fixed at 0.0005, and the number of training epochs is set to 100. The numbers of superpixels segmented for the River, Farmland, and USA datasets are 9000, 10 000, and 11 000, respectively. For each experiment, 90% of the sampling rate is used for training, the rest 10% of the sampling rate is used for validation. Use OA and Kappa coefficients to quantitatively assess the detection precision of each detection method. The higher the OA, the more accurate the detection for each class. The higher the Kappa, the more consistent the detection result map is with the GT map. The OA and Kappa coefficients are calculated as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{12}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-\mathrm{Pre}}{1-\mathrm{Pre}} \tag{13}$$

where TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively. Among them, TP represents the number of correctly predicted changed pixels, TN represents the number of correctly predicted unchanged pixels, FP represents the number of wrongly predicted changed pixels, and FN represents the number of wrongly predicted unchanged pixels. The intermediate value of Pre is calculated as follows:

$$\mathrm{Pre} = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{14}$$

B. Low Sampling Rate Experiments

All three datasets are set to a sampling rate of 1% under the present experimental conditions and compared with a series of recently published state-of-the-art methods in both subjective and objective aspects. The comparison methods include MSDFFN [2], ML-EDAN [4], MSSGU [13], CSDBF [16], SpectralFormer [32], SSFTT [33], CSANet [34], CNCMN [35], and DA-Former [36].

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

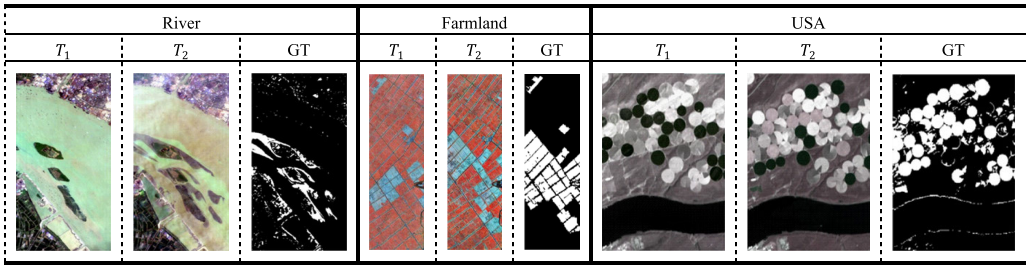


Fig. 4. Pseudo-color visualization of the datasets.

TABLE II
QUANTITY METRICS FOR THE DATASETS WITH 1% SAMPLING RATE

Year	Method	River		Farmland		USA	
		OA	$Kappa$	OA	$Kappa$	OA	$Kappa$
2021	SpectralFormer	0.9678	0.7941	0.9681	0.9230	0.9498	0.8559
2021	MSSGU	0.9622	0.7894	0.9674	0.9211	0.9445	0.8443
2022	CSDBF	0.9597	0.7498	0.9663	0.9185	0.9434	0.8436
2022	ML-EDAN	0.9536	0.6838	0.9679	0.9223	0.9389	0.8183
2022	CSANet	0.9520	0.6811	0.9650	0.9150	0.9386	0.8222
2022	SSFTT	0.9586	0.7375	0.9546	0.8903	0.9340	0.8279
2022	CNCMN	0.9576	0.7298	0.9525	0.8839	0.9363	0.8177
2023	MSDFFN	0.9587	0.6974	0.9637	0.9130	0.9487	0.8470
2023	DA-Former	0.9122	0.6031	0.9640	0.9145	0.9392	0.8354
	GTransCD	**0.9749** (↑0.71%)	**0.8366** (↑4.25%)	**0.9791** (↑1.10%)	**0.9492** (↑2.62%)	**0.9597** (↑0.99%)	**0.8810** (↑2.51%)

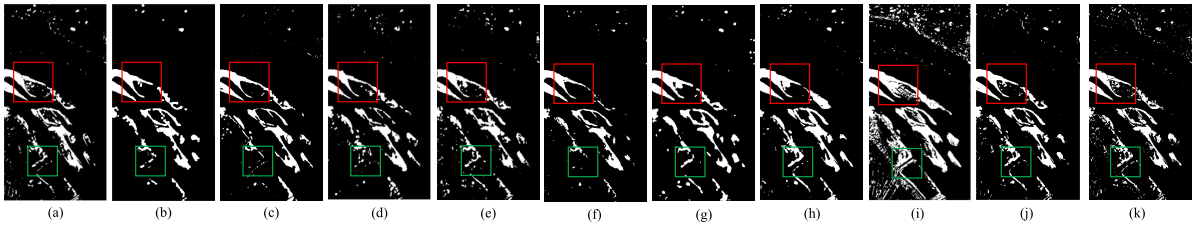


Fig. 5. CD results of different methods on the River dataset with 1% sampling rate. (a) SpectralFormer. (b) CSDBF. (c) ML-EDAN. (d) CSANet. (e) SSFTT. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

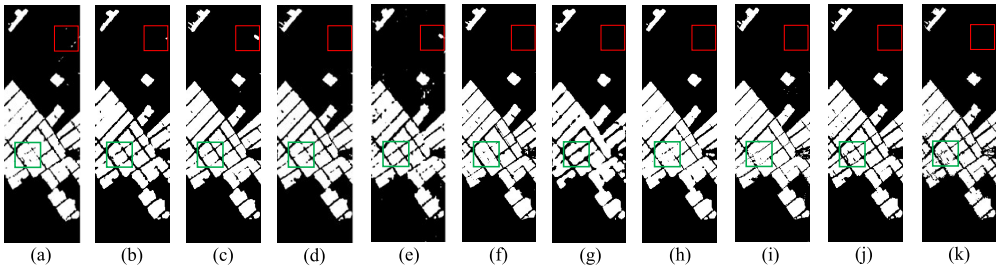


Fig. 6. CD results of different methods on the Farmland dataset with 1% sampling rate. (a) SpectralFormer. (b) CSDBF. (c) ML-EDAN. (d) CSANet. (e) SSFTT. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

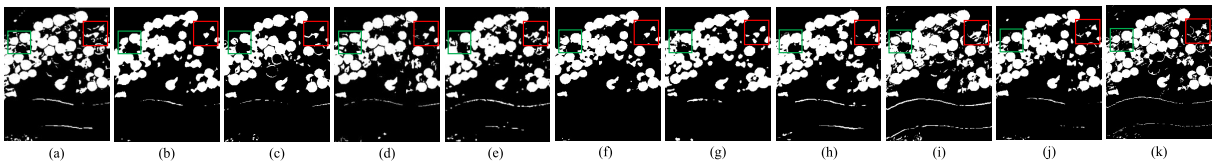


Fig. 7. CD results of different methods on the USA dataset with 1% sampling rate. (a) SpectralFormer. (b) CSDBF. (c) ML-EDAN. (d) CSANet. (e) SSFTT. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

The quantitative metrics of the detection results are compared in Table II. It is obvious that GTransCD obtains better detection results. Among them, the OA of each method on the River dataset reaches over 91%, which can meet the demand for coarse detection requirements, while the differences between Kappa values are relatively large, the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITY METRICS FOR THE DATASETS WITH NORMAL SAMPLING RATE

Year	Method	River		Farmland		USA	
		OA	$Kappa$	OA	$Kappa$	OA	$Kappa$
2021	SSCNN-S	0.9640	0.7431	0.9774	0.9440	0.9651	0.8918
2021	MSSGU	0.9735	0.8304	0.9750	0.9399	0.9649	0.8993
2022	MP-ConvLSTM	0.9750	0.8443	0.9889	0.9723	0.9677	0.8998
2022	SFBS-FFGNET	0.9670	0.7710	0.9760	0.9430	-	-
2022	CSDBF	0.9697	0.7982	0.9843	0.9620	0.9687	0.9093
2022	ML-EDAN	0.9633	0.7531	0.9861	0.9662	0.9679	0.9063
2022	CNCMN	0.9725	0.8261	0.9576	0.8977	0.9568	0.8745
2023	MSDFFN	0.9626	0.7560	0.9822	0.9569	0.9670	0.9052
2023	DA-Former	0.9389	0.6996	0.9750	0.9402	0.9626	0.8966
	GTransCD	**0.9775** (↑0.25%)	**0.8519** (↑0.76%)	**0.9898** (↑0.09%)	**0.9752** (↑0.29%)	**0.9701** (↑0.14%)	**0.9142** (↑0.49%)

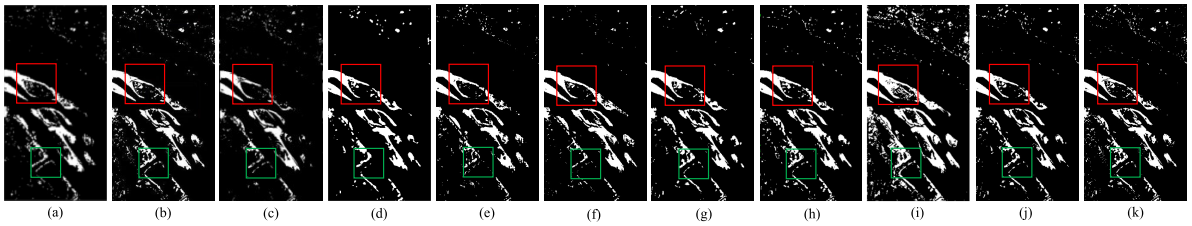


Fig. 8. CD results of different methods on the River dataset with normal sampling rate. (a) SSCNN-S. (b) MP-ConvLSTM. (c)SFBS-FFGNET. (d) CSDBF. (e) ML-EDAN. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

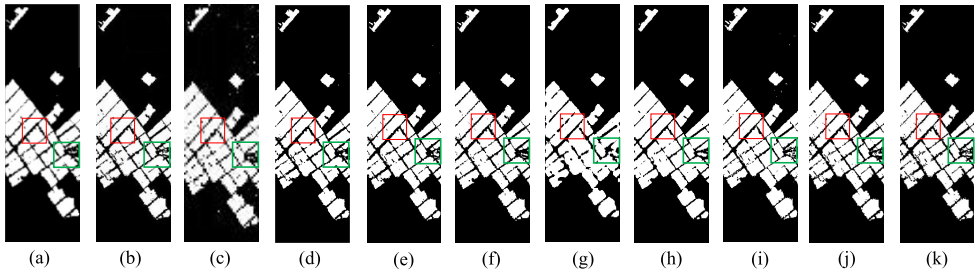


Fig. 9. CD results of different methods on the Farmland dataset with normal sampling rate. (a) SSCNN-S. (b) MP-ConvLSTM. (c) SFBS-FFGNET. (d) CSDBF. (e) ML-EDAN. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

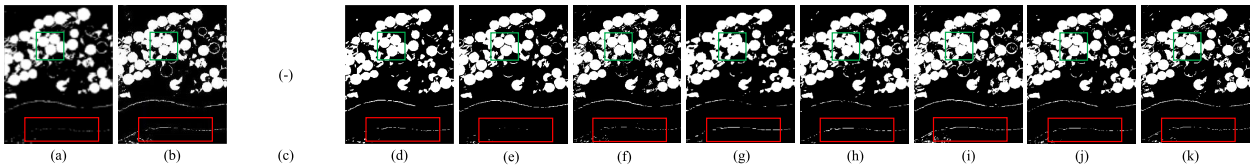


Fig. 10. CD results of different methods on the USA dataset with normal sampling rate. (a) SSCNN-S. (b) MP-ConvLSTM. (c) SFBS-FFGNET. (d) CSDBF. (e) ML-EDAN. (f) MSDFFN. (g) CNCMN. (h) MSSGU. (i) DA-Former. (j) GTransCD. (k) GT.

GTransCD in terms of Kappa values are 4.25% and 23.35% better than SpectralFormer and DA-Former, respectively, indicating that some of them are not robust enough at low sampling rates. Meanwhile, on the Farmalnd dataset, the OA values of all methods are higher than 0.95, indicating that these methods are very effective in identifying changes in blocky areas. Furthermore, GTransCD is able to achieve OA value close to 0.98, which greatly improves the detection accuracy on the Farmalnd dataset. Compared with other methods, GTransCD is also superior on the USA dataset, with a Kappa value exceeding 0.88, which is 2.51% higher than the suboptimal method, indicating that GTransCD has great advantages in experiments at low sampling rates.

1) Experiment on the River Dataset: Fig. 5 shows the visual detection result maps corresponding to the River dataset in Table II, and rectangular boxes are drawn to circle the areas with large detection differences in each method. It is evident that the detection result of CSDBF is heavily blocked, indicating that it lacks the ability to detect scattered point-shaped change areas, while ML-EDAN and MSDFFN are relatively deficient in detecting scattered points and blocks. CSANet and SSFTT have more miss alarms in the red rectangular box, while the detection result of SpectralFormer in the green

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITY METRICS FOR ABLATION EXPERIMENTS

Rate	Dataset	Index	w/o D	w/o C	w/o G	GTransCD
1%	River	OA	0.9250	0.9578	0.9675	**0.9749**
		$Kappa$	0.6425	0.7232	0.7832	**0.8366**
	Farmland	OA	0.9665	0.9568	0.9738	**0.9791**
		$Kappa$	0.9204	0.8960	0.9421	**0.9492**
	USA	OA	0.9515	0.9450	0.9548	**0.9597**
		$Kappa$	0.8606	0.8427	0.8713	**0.8810**
Norm	River	OA	0.9680	0.9606	0.9729	**0.9775**
		$Kappa$	0.8079	0.7495	0.8308	**0.8519**
	Farmland	OA	0.9846	0.9689	0.9857	**0.9898**
		$Kappa$	0.9625	0.9249	0.9672	**0.9752**
	USA	OA	0.9645	0.9563	0.9670	**0.9701**
		$Kappa$	0.8966	0.8751	0.9071	**0.9142**

rectangular box is not fine enough. CNCMN and MSSGU can detect most changes, but the results are not accuracy at the edges and small regions. The detection result of DA-Former on the River dataset is poor, many unchanged pixels are predicted into changed pixels, resulting in the low Kappa value. GTransCD has the highest consistency with GT and has excellent detection performance.

2) Experiment on the Farmland Dataset: The detection results of each method for the Farmland dataset are shown in Fig. 6. The overall detection results of all methods on the Farmland dataset are relatively accurate, but some of them are not capable of recognizing the unchanged information, resulting in noise in the background. The red box emphasizes that some methods have significant miss alarms, these methods are not strong enough to discriminate pseudo-change features at low sampling rates, while the green box highlights that GTransCD can achieve a superior result for detecting changed edges, and other methods are weak in capturing details and have more false alarms at edges.

3) Experiment on the USA Dataset: The detection results of each method for the USA dataset at a low sampling rate are shown in Fig. 7. It can be seen that GTransCD can detect most of the changed areas more accurately, and the contents in the green and red boxes highlight the high accuracy in detecting edges, which are rounder and smoother than other methods. Unfortunately, the detection of the river lines is lacking, probably due to the lack of detailed segmentation of the superpixels.

C. Normal Sampling Rate Experiments

The sampling rates of all three datasets are set to 3.36%, 20.95%, and 9.77% under the same experimental conditions and compared with a series of recently published state-of-the-art methods in both subjective and objective aspects. The comparison methods include: SSCNN-S [37], MSSGU [13], MP-ConvLSTM [3], SFBS-FFGNET [38], CSDBF [16], ML-EDAN [4], CNCMN [35], MSDFFN [2], and DA-Former [36].

The objective metrics for the detection results of each comparison method on the three HSI-CD datasets are given in Table III. The proposed GTransCD still achieves the optimal detection results, MP-ConvLSTM achieves suboptimal results on the River and Farmland datasets and CSDBF achieves suboptimal results on the USA dataset. For the River dataset, there are only a few methods that can achieve a Kappa value above 0.8 after increasing the sampling rate, indicating that some of them do not perform well in the detection of scattered changes. For the Farmland dataset, some methods can reach OA values of 0.98, but the difference between OA and Kappa is too much, such as MSDFF and CSDBF methods, suggesting that these methods may be strongly affected by class imbalance factor. For the USA dataset, only GTransCD can obtain OA above 0.97 and Kappa above 0.91, bringing the detection result of the USA dataset to a new level. Compared with other methods, CSDBF, ML-EDAN, and MSDFFN achieve higher Kappa values, which demonstrates greater consistency with the GT map.

1) Experiment on the River Dataset: Fig. 8 shows a visual comparison of each method on the River dataset. It can be clearly observed that CSDBF misses many detections of scatter-shaped change points. SSCNN-S, SFBS-FFGNET, and ML-EDAN methods have significant miss alarms in the red box, and MSDFFN has obvious miss alarms in the green box. In contrast to MSSGU and CNCMN, which achieves competitive results in detecting scatter-pattern changes, DA-Former has a high level of false alarms. MP-ConvLSTM and GTransCD have a high degree of consistency with GT subjectively, indicating that GTransCD achieves satisfactory results subjectively and objectively.

2) Experiment on the Farmland Dataset: The subjective result comparison of different methods on the Farmland dataset is shown in Fig. 9. There are some obvious misdetected pixels in the background of the SFBS-FFGNET detection result map, and the edges are messy, which is not consistent with the GT map; at the same time, there is a small amount of noise in the background of DA-Former detection result, but it performs well in terms of detection details. SSCNN-S and MSDFFN are not good at detecting the changed content in the green box, and there are a lot of miss alarms; the edges of the detection results of CSDBF and ML-EDAN are smoother than GTransCD, and lack the ability of fine detection. The capability of CNCMN and MSSGU to capture edges during change detection is weak relatively, resulting in the loss of many details, indicating that the model is not strong enough to discriminate mixing regions. MP-ConvLSTM has the best performance among the comparison algorithms, but there is still a certain gap with GTransCD in terms of consistency performance.

3) Experiment on the USA Dataset: Fig. 10 shows the change detection prediction results obtained by different methods on the USA dataset. The results show that in the red box, GTransCD obtains the optimal detection result of river lines, while SSCNN-S and ML-EDAN lack the capture of changes in this area. And the detection results of CSDBF and MSDFFN are better, even if the detection edges are continuous. Meanwhile, DA-Former still performs well in detecting changed details. The detection superiority of GTransCD can also be observed in the green box, with the highest agreement with the GT map.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

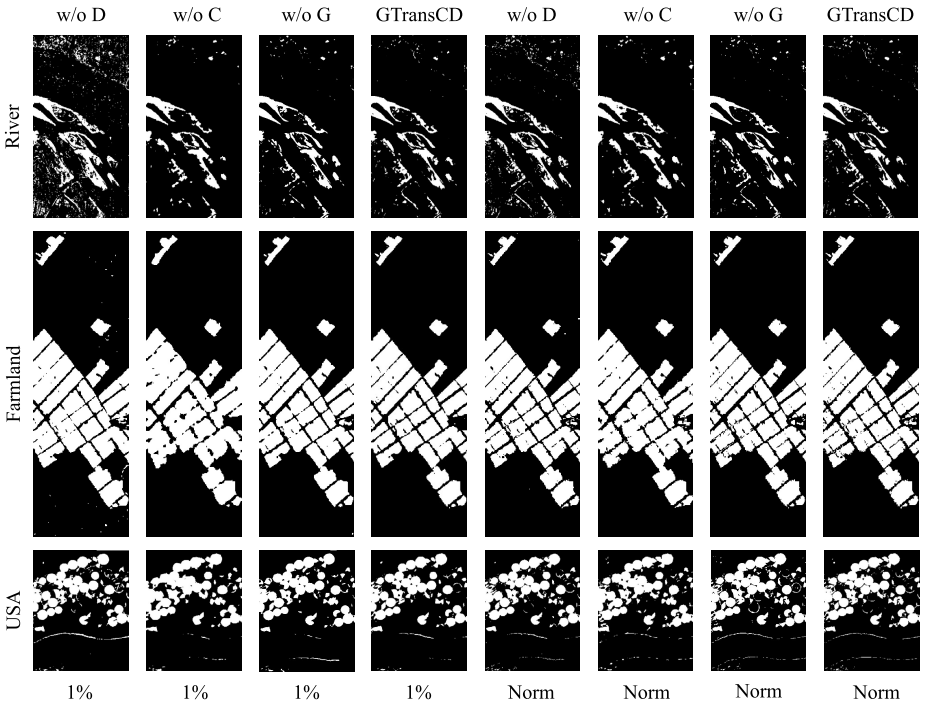


Fig. 11. CD visualize results of ablation experiments on three datasets with two sampling rates.

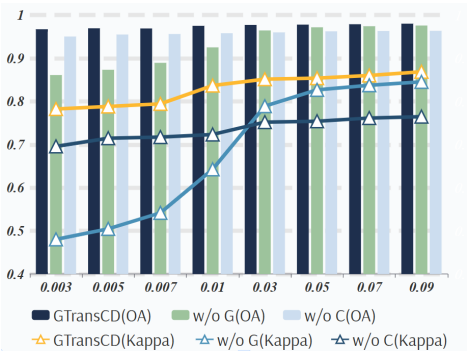


Fig. 12. CD metric results of ablation experiments on the river dataset with multiple sampling rates.

D. Ablation Experiments and Discussions

1) Ablation Experiments: GTransCD mainly consists of two proposed branches, one is the GTrans-guided differential temporal analysis branch, and the other is the original bitemporal concatenated branch. For this reason, ablation experiments are designed for these two branches, respectively defined as w/o D and w/o C experiments. It should be noted that since the GCF unit needs to receive the above two branches as input, therefore, neither GCF unit is contained in each above ablation experiment. Here, we append a w/o G experiment, which is the ablation of GCF unit. Table IV presents the comparison of the objective metrics of the ablation experiments for the three datasets under two groups of sampling rates, with the sampling rate of "Norm" following the sampling rate of the IV.C experiment.

To compare the role of each branch in GTransCD more conveniently, the specific visual detection results of the three datasets are given in Fig. 11. It is clearly observed that at 1% sampling rate, the River result map of the w/o D experiment has a large number of false alarms, while the w/o C experiment has an error accumulation effect due to superpixel segmentation, and the ability to capture change details needs to be improved. At this time, GTransCD can realize the organic combination of these two branches, using pixel-level features to correct the wrongly segmented superpixel-level features. When the sampling rate comes to "Norm," the detection result of the w/o C experiment is lower than that of the w/o D experiment is still because considering the calculation cost and efficiency, the number of superpixels is not large enough, which increases the accumulation of errors.

2) Discussion on Sampling Rates: To further show how each branch performs at different sampling rates, the results of the ablation experiments on the River dataset at sampling rates of 0.3%, 0.5%, 0.7%, 1%, 3%, 5%, 7%, and 9% are presented in Fig. 12, where the OA values are given as bars and the Kappa values are given as lines.

It is shown that the w/o C is significantly better than the experimental results for w/o D under extremely low sampling conditions, and the difference in Kappa values is particularly obvious, here, the pixel-level processing mode is much less effective than the superpixel-level processing mode, indicating that under low sampling rates condition, information from labeled samples can be effectively propagated to unlabeled samples in the form of superpixels through the global message passing mode. As the sampling rate gradually increases, the experimental results for w/o D outperform those for w/o C, indicating that the superpixel-based graph modeling method is no longer applicable to the River dataset under high sampling rates condition, with a degree of negative impact due to the accumulated errors in superpixel segmentation,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

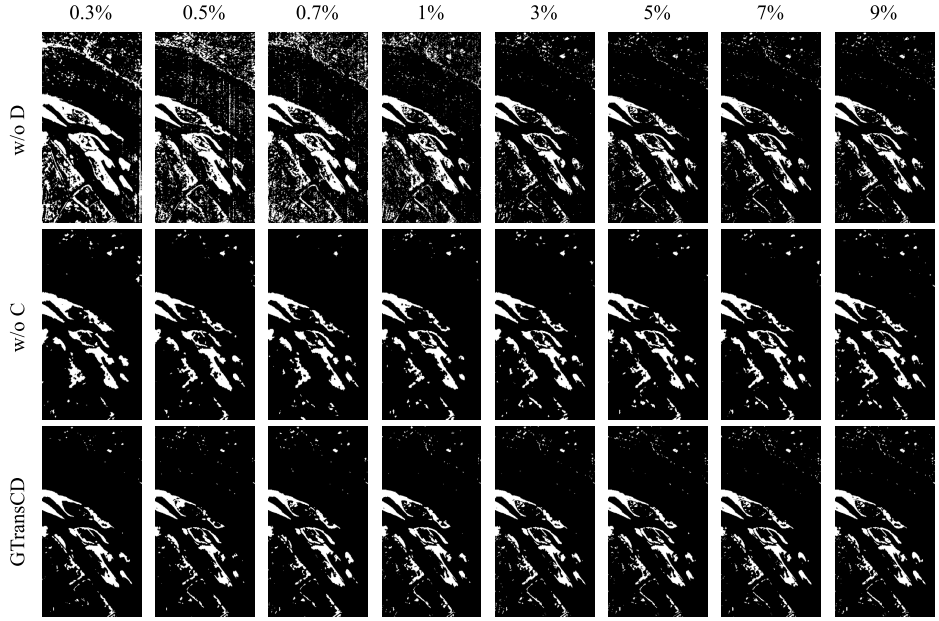


Fig. 13. CD visualize results of ablation experiments on the River dataset with multiple sampling rates.

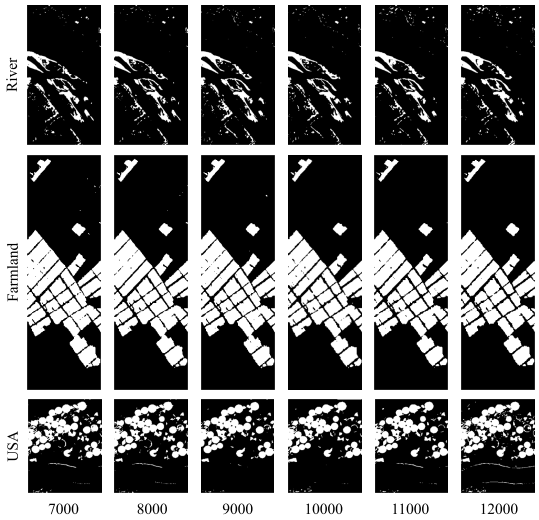


Fig. 14. CD visualize results of GTransCD on each dataset with multiple superpixel-level graphs.

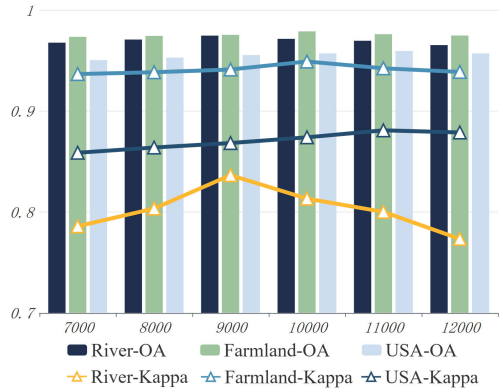


Fig. 15. CD metric results of GTransCD on each dataset with multiple superpixel-level graphs.

whereas pixel-level processing is more advantageous under the condition of sufficient samples.

Fig. 13 shows the visualization results corresponding to Fig. 12, that is the change detection results of the ablation experiment on the River dataset at various sampling rates. When the sampling rate is less than 1%, there is a large amount of salt and pepper noise in the w/o D experiment result map, which detects the unchanged pixels as the changed pixels. The above false alarms occur due to the small number of training samples that prevent the network from distinguishing the difference between the real change information and the pseudo-change information. In contrast to the w/o C experiment, due to superpixel segmentation, it loses the recognition of small changes, causing certain miss alarms occur, but can capture most of the real change objects. GTransCD organically combines the two branches, which not only corrects a large number of false alarms in the w/o D experiment but also supplements the miss alarms in the w/o C experiment. When the sampling rate comes to 3%, the detection results of the w/o D experiment outperform w/o C experiment, because the effective capture of pixel-level change information makes it possible to identify the real change information, but is prone to misdetection at the edges, which then needs to be supplemented by the w/o D branch. This shows that the w/o C experiment is less dependent on sample rates and can still play a good detection performance when there is not enough learnable data. While the pixel-level w/o D experiment enables the network to make effective judgments for each pixel when the samples are rich. Under multiple sampling rates, different branches play their respective advantages, so that GTransCD can achieve the best subjective and objective detection results.

3) Discussion on the Number of Superpixels: As above described, it is necessary to construct a superpixel-level

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

graph based on superpixel segmentation, so that, the number of superpixels is a hyperparameter worth discussing, which affects the size of the graph, which means that it affects the computing performance directly. That is, the smaller the segmentation scale, the larger the number of superpixel subregions, the larger the constructed superpixel-level graph, the higher the computational complexity; Conversely, the larger the segmentation scale, the smaller the number of superpixel subregions, the smaller the constructed superpixel-level graph, and the lower the computational complexity. Figs. 14 and 15 shows how each dataset behaves on multiple superpixel-level graphs at 1% sampling rate.

It can be seen that when the segmentation scale is small, the size of each subregion is large, underclassification occurs from time to time, the misclassification pixels that need to be corrected by pixel-level branch are overmuch, so that there are more noises in detection results. When the segmentation scale is large, the size of each subregion is small, overclassification occurs from time to time, the current superpixel-level branch aggregates too much invalid background information, which also causes noises. Hence, according to the experiments, the number of superpixels should be set based on the characteristics of each dataset.

V. Conclusion

In this article, a GTrans-guided multitemporal information united framework is proposed for HSI-CD in view of the problems of few samples and inadequate information perception of distant ground objects in existing HSI-CD frameworks. The framework organically blends the change features guided by salient relationship strengthening GTrans module with the original bitemporal concatenated features, and is able to propagate labels to unlabeled samples in a semi-supervised learning mode by constructing the topological structured graph, reducing the reliance of the network on a large number of training samples. In particular, the proposed salient relationship strengthening GTrans module applies the transformer to a graph with superpixels as nodes, aiming to capture change information in the whole image under the condition of low-computational cost. It breaks through the traditional definition of adjacency relationship and obtains the affinity relationship between any two sub-regions. Furthermore, the GCF unit effectively injects the change features guided by GTrans module into the original bitemporal concatenated features, realizing the blend of multitemporal information. The superior detection performance of GTransCD is demonstrated by multi-sampling rate experiments on three commonly used HSI-CD datasets, and the comparison of recent experiments verifies that GTransCD outperforms existing SOTA methods.

References

[1] S. Liu, D. Marinelli, L. Bruzzone, and F. Bovolo, "A review of change detection in multitemporal hyperspectral images: Current techniques, applications, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 140–158, Jun. 2019.

[2] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[3] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529816.

[4] J. Qu, S. Hou, W. Dong, Y. Li, and W. Xie, "A multilevel encoder–decoder attention network for change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518113.

[5] A. Song, J. Choi, Y. Han, and Y. Kim, "Change detection in hyperspectral images using recurrent 3D fully convolutional networks," *Remote Sens.*, vol. 10, no. 11, p. 1827, Nov. 2018.

[6] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial-channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402214.

[7] C. Liang and Z. Chen, "A self-supervised hierarchical clustering network for multiple change detection in multitemporal hyperspectral images," in *Proc. 12th Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2022, pp. 1–4.

[8] J. Sun, J. Liu, and L. Xiao, "Intrinsic decomposition model-guided two-stream coupled autoencoder for unsupervised hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[9] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A comprehensive survey on graph neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 1, pp. 4–24, 2021, doi: 10.1109/TNNLS.2020.2978386.

[10] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[11] L. Mou, X. Lu, X. Li, and X. X. Zhu, "Nonlocal graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8246–8257, Dec. 2020.

[12] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[13] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "Multilevel superpixel structured graph U-Nets for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516115.

[14] W. Guo, G. Xu, W. Liu, B. Liu, and Y. Wang, "CNN-combined graph residual network with multilevel feature fusion for hyperspectral image classification," *IET Comput. Vis.*, vol. 15, no. 8, pp. 592–607, Dec. 2021.

[15] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[16] X. Wang, K. Zhao, X. Zhao, and S. Li, "CSDBF: Dual-branch framework based on temporal–spatial joint graph attention with complement strategy for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540118.

[17] J. Qu, Y. Xu, W. Dong, Y. Li, and Q. Du, "Dual-branch difference amplification graph convolutional network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5519912.

[18] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[19] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[20] A. Shafique, S. T. Seydi, T. Alipour-Fard, G. Cao, and D. Yang, "SSViT-HCD: A spatial–spectral convolutional vision transformer for hyperspectral change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6487–6504, 2023, doi: 10.1109/JSTARS.2023.3251646.

[21] Y. Wang et al., "Spectral–spatial–temporal transformers for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5536814.

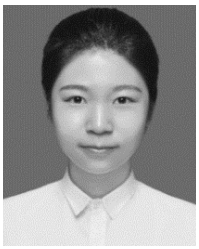
[22] J. Ding, X. Li, and L. Zhao, "CDFormer: A hyperspectral image change detection method based on transformer encoders," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[23] M. Hu, C. Wu, and B. Du, "EMS-Net: Efficient multitemporal self-attention for hyperspectral change detection," 2023, *arXiv:2303.13753*.

[24] B. Sun, Q. Liu, N. Yuan, J. Tan, X. Gao, and T. Yu, "Spectral token guidance transformer for multisource images change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 2559–2572, 2023.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.

[25] Y. Huang, L. Zhang, C. Huang, W. Qi, and R. Song, "Parallel spectral–spatial attention network with feature redistribution loss for hyperspectral change detection," *Remote Sens.*, vol. 15, no. 1, p. 246, Dec. 2022.
[26] X. Wang, K. Zhao, X. Zhao, and S. Li, "TriTF: A triplet transformer framework based on parents and brother attention for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507213.
[27] M. Hu, C. Wu, and L. Zhang, "GlobalMind: Global multi-head interactive self-attention network for hyperspectral change detection," 2023, *arXiv:2304.08687*.
[28] X. Wei, Q. Yang, Y. Gong, N. Ahuja, and M.-H. Yang, "Superpixel hierarchy," *IEEE Trans. Image Process.*, vol. 27, no. 10, pp. 4838–4849, Oct. 2018.
[29] Q. Wang, Z. Yuan, Q. Du, and X. Li, "GETNET: A general end-to-end 2-D CNN framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 3–13, Jan. 2019.
[30] Y. Yuan, H. Lv, and X. Lu, "Semi-supervised change detection method for multitemporal hyperspectral images," *Neurocomputing*, vol. 148, pp. 363–375, Jan. 2015.
[31] M. Hasanlou and S. T. Seydi, "Hyperspectral change detection: An experimental comparative study," *Int. J. Remote Sens.*, vol. 39, no. 20, pp. 7029–7083, Oct. 2018.
[32] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.
[33] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.
[34] R. Song, W. Ni, W. Cheng, and X. Wang, "CSANet: Cross-temporal interaction symmetric attention network for hyperspectral image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[35] Q. Liu, L. Xiao, N. Huang, and J. Tang, "Composite neighbor-aware convolutional metric networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1–15, 2022, doi: 10.1109/TNNLS.2022.3232532.
[36] Y. Wang, J. Sha, L. Gao, Y. Zhang, X. Rong, and C. Zhang, "A semi-supervised domain alignment transformer for hyperspectral images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525411.
[37] T. Zhan et al., "SSCNN-S: A spectral–spatial convolution neural network with Siamese architecture for change detection," *Remote Sens.*, vol. 13, no. 5, p. 895, Feb. 2021.
[38] X. Ou, L. Liu, B. Tu, G. Zhang, and Z. Xu, "A CNN framework with slow-fast band selection and feature fusion grouping for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524716.



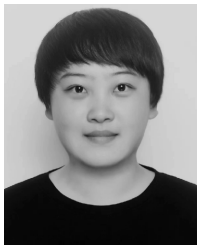
Xiaoyang Zhao was born in 1994. She received the M.S. degree with the School of Computer and Artificial Intelligence, Liaoning Normal University, Dalian, China, in 2019, where she is currently pursuing the Ph.D. degree with the School of Geographical Sciences.

Her research interests include remote sensing image processing and deep learning.



Siyao Li was born in 1999. She is pursuing the M.S. degree with the School of Computer and Artificial Intelligence with Liaoning Normal University, Dalian, China.

Her research interests include remote sensing image processing and deep learning.



Tingting Geng was born in 2000. She is pursuing the M.S. degree with the School of Computer and Artificial Intelligence with Liaoning Normal University, Dalian, China.

Her research interests include remote sensing image processing and deep learning.



Xianghai Wang was born in 1965. He received the Ph.D. degree from Jilin University, Changchun, China, in 1999.

He is currently a Professor with the School of Geographical Sciences and the School of Computer and Artificial Intelligence, Liaoning Normal University, Dalian, China. He is a Senior Member of the China Computer Federation (CCF), Beijing, China. His research interests include computer graphics, multimedia information processing, and multi-band remote sensing image information processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 12:35:48 UTC from IEEE Xplore. Restrictions apply.